

Improved Stock Market Structure Using Cryptography

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Abstract. The stock market has two primary functions, that of providing liquidity and price discovery. While historically, the market micro-structure was mostly ignored or assumed to function ideally for the purpose of asset pricing, O'Hara (Journal of Finance, 2003) established that both liquidity and price discovery affect asset pricing, and consequently asset returns. In this work, we extend the analysis of Easley and O'Hara (Journal of finance, 2004) to study common stock market mechanisms, including open-bid and sealed-bid auctions, as well as the fact that the auctioneer may not be trustworthy. Our analysis shows that open-bid auctions dis-incentivize stock research, leading to higher volatility. On the other hand, we also show that in sealed-bid auctions, the price and volume of an auction reveals only partial private research information, thus retaining incentive for traders to do research. We also show how secure computation can be used to build auctions with a distributed auctioneer, where the clearing and partial-fulfillment information of the auction does not reveal unfilled orders of others, thus maintaining private information.

1 Introduction

The advent of bitcoin and other cryptocurrencies has undoubtedly demonstrated that cryptography can lead to disruptive technologies. In this work we propose that modern cryptography can also lead to enhanced stock markets that incentivizes investors to conduct independent research in a firm's stock and/or stock sector and thus lead to a better overall information structure surrounding the stock. In a ground-breaking work, Easley and O'Hara [EO04] demonstrated a rational expectations equilibrium model that *incorporates private information* to show that the quantity and quality of information lowers market risk and hence affects asset prices. We will use the same model along with additional analysis of practical mechanisms such as clearing price double auctions (CPDA), to reinforce the importance of incentivizing independent research in a stock. Further, we propose the use of modern cryptography, in particular secure multi-party computation [Yao86,GMW87,BGW88,CCD88,BCD⁺09] to replace open order-books

and/or real-world auctioneers with *ideal* auctioneers, so that the information garnered by this costly independent research is not revealed to uninformed traders, and hence preserving the incentive to conduct such independent research. The ideal auctioneer is realized using a distributed computing system, split between various entities (possibly including private banks/exchanges and regulators), so that no collusion of a subset of these entities can infer any useful information about the order book.

We first need to establish, in an economic-theory sense, that sealed-bid auctions are the correct mechanism for the stock-market auctions. For otherwise, one could have an open order-book system, and the clearing mechanism itself becomes transparent and there is no need for secure computation by a distributed computing system. To this end, we model two kinds of traders, *informed traders* and *uninformed traders*. The former have put in an effort, sometimes at a cost, to do research on the stock, whereas the latter trade purely based on market technicals. We model every trader to have a common information structure regarding a stock's future price v , e.g. a normal distribution with a known mean $\bar{v}$ and precision³ ρ . In addition, the informed traders have some research-based information about the stock's future price, such as customer surveys, supply-chain analysis, or even scientific knowledge. Easley and O'Hara in a ground-breaking work [EO04] modeled this additional information itself as a signal s which takes a value that is the actual future price v plus some noise that is sampled from a normal distribution (with mean zero and a certain precision). Then, in the rational expectations equilibrium model or the Walrasian equilibrium [Lev06], they show that equilibrium price only partially reveals the signal s . For readers not familiar with Walrasian equilibrium, it is an information-theoretic equilibrium (i.e without involving any mechanism) where a redistribution of stocks is in equilibrium with prices of various stocks, with each trader maximizing their individual utility while respecting their budget. We describe this model in more detail in Section 4, but for now it suffices to emphasize that this equilibrium is obtained where each trader has their own personal valuation/utility of a stock (e.g. the informed traders may base it on s) and the only information available to the uninformed is the equilibrium prices and other public statistics.

Interestingly, this "non-effective" equilibrium (with some further refinement) can already be used to show, in an informal sense, that open order-book auction leads to more volatility in the stock market, which can be construed as mis-allocation of capital. To see this, suppose that the signal s obtained (by conducting research) by informed traders is given for free to uninformed traders. Then, it is as good as assuming that all traders receive the signal s (which is more pointed information about the future price than the a priori mean $\bar{v}$), *except* that some traders, the informed traders, incurred an extra cost to conduct the research. First, ignore this extra cost, and let's calculate the return obtained by Easley-O'Hara Walrasian equilibrium. The good news is that with more research conducted by traders, the premium demanded by the traders to hold the stock is lower. However, everyone is getting the same return as all have the

³ Precision is reciprocal of variance.

same signal s . But, if we include the extra cost incurred by informed traders, they are getting a worse return. Hence, in equilibrium they must stop doing research (this argument is due to Grossman and Stiglitz [GS80]). But then, no one gets the signal s , and must operate based on the a priori future mean price $\bar{v}$, which of course leads to traders demanding more premium to hold the risky stock (where traders have very low information about it). The only reason this argument is informal is that we are not talking about a mechanism here, but an information-theoretic equilibrium. For example, a mechanism where traders submit their open bids, potentially, may not reveal signal s to uninformed traders.

To make this argument more formal, in this work we extend the analysis of Easley and O’Hara to actual mechanisms, such as clearing price double auctions (CPDA). It is worth remarking here that the stock market with its repeated auctions (or continuous auctions) is essentially trying to replicate the phenomenon known as Walras’ tatonnement, an almost end-less trial and error approach to “convergent” Walrasian equilibrium price-discovery; see e.g. [Lev06]. However, in the process, even for sealed-bid auctions, it discloses price and volume information to all traders of all intermediate auctions. Thus, our task is two fold. First, we need to show that open-bids do reveal the signal of informed traders. This is rather easy to demonstrate. Second, we need to show that even with price *and* volume information given to all traders, it only partially reveals an informed trader’s signal s . Indeed, we show in Section 4.1 that in a Nash equilibrium, the price and volume only partially reveal a traders signal, as long as the orders are sealed (i.e. hidden or encrypted).

A final desideratum is to design the clearing price auction mechanism in such a way that partial fulfillment of orders of individual traders, do not reveal to such traders the (sealed) order-book, i.e. orders of others that were not filled. We mention that there are certain variants of CPDA such as uniform clearing or discriminatory buyer’s-bid, which can potentially reveal the order-book (on revelation of partial fulfillment). Yet another important requirement is the involvement to incorporate market orders. We show how to satisfy these requirements by describing the precise mechanism as an ideal functionality in Section 2.

1.1 Related Work

While secure multi-party computation (MPC) [Yao86,GMW87,BGW88,CCD88] has been adequately demonstrated to be useful in replacing auctioneers in clearing price auctions [BCD⁺09], no economic justification of using sealed-bid auctions and secure computation was given in [BCD⁺09] beyond mentioning that the auctioneer is assumed to be ideal in economic modeling of auctions and equilibrium theories.

In economic literature, sealed bid auctions for one-sided auction have already been demonstrated to be better than open-bid auctions [ALS11] in the US timber auctions, although efficiency vs revenue gains are in conflict in that analysis. In commodities exchange, a mechanism is called “efficient” if it ends up in possession of one who values it most [Mas00]. In a one-sided auction, such as timber

auctions, there is only one seller but many buyers. However, two-sided auctions are notoriously more difficult to analyze.

For two-sided auctions, the best analysis known is that if all investors are equally informed then in a sealed-bid double auction [SW89] the clearing price rapidly converges to the rational expectations equilibrium price. Further, in the individual private belief model it is shown that if the number of traders is large then they converge to bidding truthfully, i.e. in accordance with their private valuation. Finally, when it comes to bidding strategies or allowed schemes, [KN04] have shown that discrete bidding strategies (as opposed to continuous bidding functions) make the clearing price auction efficient in the *one-sided auction*.

1.2 Organization

The rest of the paper is organized as follows. In Section 2 we describe the specific CPDA auction we propose as stock market mechanism. In Section 3, we briefly cover economic and game theory basics. The rational expectations equilibrium analysis of Easley and O'Hara in presence of private information [EO04] is reviewed in Section 4. Finally, the effects of micro-structure of our CPDA as it pertains to semi-disclosure of private information is proved in Sections 4.1. We end with conclusion and implications in Section 5.

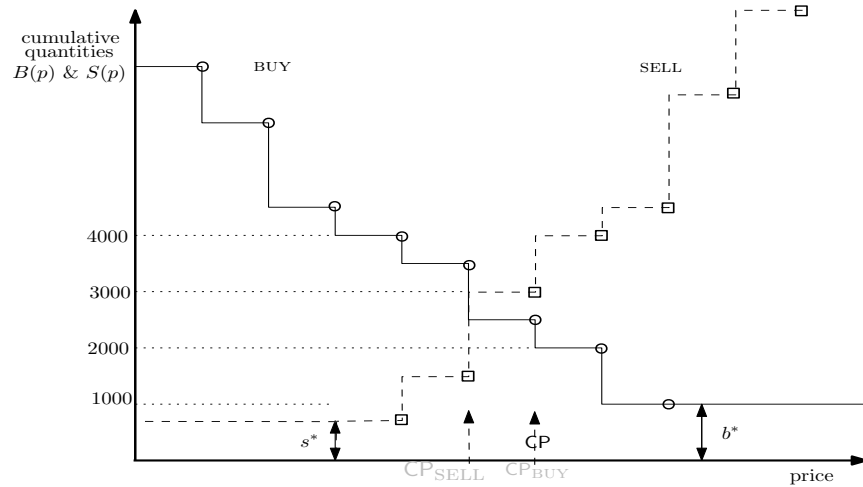


Fig. 1. CPDA Clearing Mechanism

2 Stock Market Mechanism

We first describe the CPDA mechanism assuming an ideal functionality $\mathcal{F}_{\text{CPDA}}$ or a single ideal auctioneer. In this description, the investors do not encrypt

their inputs, as the ideal auctioneer is by definition trusted to not leak this information to anyone. Once the single ideal auctioneer is precisely described, especially as to what its inputs and outputs are and how the output is computed from the inputs, we can then move on to describe how the inputs are encrypted and how this ideal auctioneer can be replaced and implemented using secure multi-party-computation by a set of independent auctioneers.

An ideal functionality takes inputs from all the investors (or their brokers), computes the required function(s) and returns the specified outputs to the individual investors and public. No other information about the different investor's input is leaked to any of the investors or observers. Since our aim is to hide the investors' bid-ask size and price, especially those that were not matched or allotted, even in this ideal description we have to pay extra attention to whether partial fulfillment ratios can leak information about unallocated orders. For an example of the following CPDA mechanism, see Fig. 1.

Ideal Functionality $\mathcal{F}_{\text{CPDA}}$ (CPDA Mechanism)

- All brokers submit their clients' bids and asks as follows:
 - Each broker is pre-assigned a broker ID, say `BROKER-ID`.
 - Each client can make two kinds of bids and two kinds of asks. One of the two kinds in either case is *market price*.
 - If it is not a market price bid or ask, the bid or ask is a list of pairs $\langle z_i, p_i \rangle$, where z_i is the quantity bid to buy at price p_i (or quantity z_i to sell with ask-ing price p_i). Each price shall be in (positive integer) cents.
 - If it is market-price the list of pairs is a singleton with the pair $\langle z, * \rangle$.
 - Thus, each client's order will be specified as (order-type = market/limit, buy/sell, list-of-pairs).
 - A client can submit multiple orders.
 - The broker assigns a random ID to each order (large enough to be unique), say `ID`, and submits (`BROKER-ID`, `ID`, order-type = market/limit, buy/sell, list-of-pairs).
- The ideal functionality $\mathcal{F}$ computes the following:
 1. $\mathcal{F}$ outputs the total number of buy and sell orders to the public.
 2. For each possible price p , $\mathcal{F}$ computes $b(p)$ as the sum of all z , where the sum is over all brokers and all of their orders marked `BUY` such that its corresponding list-of-pairs contains a pair $\langle z, p \rangle$.
 3. $\mathcal{F}$ also computes b^* as the sum of all z , where the sum is over all brokers and all of their orders marked `BUY` such that its corresponding list-of-pairs contains a pair $\langle z, * \rangle$.
 4. Similarly using the orders marked `SELL`, $\mathcal{F}$ computes $s(p)$ for each possible price p , as well as s^* .
 5. For each p , let $B(p) = b^* + \sum_{p' \geq p} b(p')$, and $S(p) = s^* + \sum_{p' \leq p} s(p')$.
 6. Next, $\mathcal{F}$ computes the least price p , denoted cp_{BUY} , such that $B(p) \leq S(p)$.

7. $\mathcal{F}$ also computes the largest price p , denoted cp_{SELL} , such that $B(p) \geq S(p)$.
8. If $S(\text{cp}_{\text{SELL}}) > B(\text{cp}_{\text{BUY}})$, then clearing price $\text{cp} = \text{cp}_{\text{SELL}}$, otherwise $\text{cp} = \text{cp}_{\text{BUY}}$. Also, $\mathcal{F}$ outputs cp to the public.
9. All buy orders which are either market price or if their bidding price is greater than or equal to cp_{BUY} are given full allotment at price cp . $\mathcal{F}$ outputs all such bids, along with their BROKER-ID and ID to the respective brokers..
10. Similarly, all sell orders which are either market price or if their asking price is less than or equal to cp_{SELL} are given full allotment at price cp . $\mathcal{F}$ outputs all such asks along with their BROKER-ID and ID to the respective brokers.
11. • If $\text{cp} = \text{cp}_{\text{BUY}}$, $\mathcal{F}$ sorts all sell orders with limit price cp_{BUY} by decreasing order size (i.e. z_i in the asks). Then it outputs the sell orders starting with the largest, as long as the sum total remains less than $B(\text{cp}_{\text{BUY}}) - S(\text{cp}_{\text{SELL}})$. The last sell order maybe partially filled, so that the sum total is exactly $B(\text{cp}_{\text{BUY}}) - S(\text{cp}_{\text{SELL}})$.
 • Else, $\mathcal{F}$ sorts all buy orders with limit price cp_{SELL} by decreasing order size (i.e. z_i in the bids). Then it outputs the buy orders starting with the largest, as long as the sum total remains less than $S(\text{cp}_{\text{SELL}}) - B(\text{cp}_{\text{BUY}})$. The last buy order maybe partially filled, so that the sum total is exactly $S(\text{cp}_{\text{SELL}}) - B(\text{cp}_{\text{BUY}})$.
12. $\mathcal{F}$ outputs the total volume of trades to the public.

Remark. The reason why the partial fulfillment in step 11 is prioritized by size of the order and not performed on a random basis is because if the partial fulfillment was based on randomization then a curious investor can learn the size of the gap $B(\text{cp}_{\text{BUY}}) - S(\text{cp}_{\text{SELL}})$ by submitting many different small orders and noting the fraction of those orders that get allotted. Then, comparing it with the total volume which is public information, he/she can infer $B(\text{cp}_{\text{BUY}}) - S(\text{cp}_{\text{SELL}})$. Hence, it is important to give precedence to larger orders at clearing price.

Note that it may be rather inefficient to realize an ideal functionality (by MPC) that sorts the orders (by secret quantities z_i) as described above in step 11. So, we suggest an alternate randomized but more efficient mechanism here, which statistically has the same effect of hiding the difference $B(\text{cp}_{\text{BUY}}) - S(\text{cp}_{\text{SELL}})$. We will describe only the case for $\text{cp} = \text{cp}_{\text{BUY}}$ as the other case is symmetric. The step 11 is then alternatively given by the following algorithm:

Set $d := s(\text{cp}_{\text{BUY}})$. Set $\psi := B(\text{cp}_{\text{BUY}}) - S(\text{cp}_{\text{SELL}})$.
 Let L be the list of sell orders with limit selling price cp_{BUY} (ordered randomly).
 while ($d \geq 1$) do
 for all orders i in L that are not marked filled *and* (order-size $z_i > d$) do
 if ($z_i \geq \psi$) then output order BROKER-ID, ID with allotment ψ . Exit;
 else output order BROKER-ID, ID with allotment z_i and mark it filled.
 $\psi := \psi - z_i$.

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    (end for loop)
   $d := d/2$ 
(end while loop)

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Readers not familiar with secure multi-party computation are referred to "Secure MPC Goes Live" [BCD⁺09] or [CHK⁺12]. These works describe different, but now standard, ways of implementing the ideal functionality using a distributed set of parties (henceforth called servers). In particular, the former is built on [BGW88, CCD88] and the latter uses Rabin's Oblivious Transfer (OT) [JH85] and is built on [GMW87]. Both these protocols are in the semi-honest security model, which means that the servers are assumed to not deviate from the protocol, but are allowed to collect any information they can from the execution of the protocol. There has also been a big trend in offline-online MPC, where the offline stage performs pre-computation to populate the servers with correlated randomness, and when the actual inputs arrive, the online stage can run much more efficiently and using very simple information-theoretic primitives [Bea91]. In fact, in this paradigm, the SPDZ protocol [DPSZ12], even offers security against $n - 1$ malicious servers, where n is the number of servers running the distributed auctioneer. This protocol however is not fault-tolerant like [BCD⁺09] or more general fault-tolerant secure MPC schemes like [BGW88].

An important aspect of implementation is how the traders share their bids/asks to the servers, and a dishonest trader/broker can potentially submit a malformed set of shares that can derail the eventual clearing price computation. However, this is resolved using well-known techniques such as verifiable secret-sharing [BGW88] which can also withstand a $< n/3$ corrupt and malicious servers.

Note that in the mechanism above, we assume that brokers run the protocol through $\mathcal{F}$. However, in the real world there can be a large number of brokers (from different countries, as well). Thus, to be practical, we will assume that some small subset of these brokers, say four, and a regulating authority such as the securities and exchange commission (SEC) actually run the protocol emulating and replacing $\mathcal{F}$. In standard cryptography parlance this would then be called a secure 5-PC (five-party computation). Recall, different known 5-PC protocols give different security guarantees against collusion. We recommend using a 5-pc protocol that requires all five parties to collude before they can get any useful information, especially if a watchdog agency is one of the five participating auctioneers.

The brokers would still need to encrypt their bids and asks (for all of their clients). For some versions of MPC, the brokers may instead be just required to implement splitting their inputs using known simple secret sharing techniques and then sending the shares (using a cryptographically secure channel, e.g. SSL) to the five parties implementing MPC (the secret sharing can be a basic additive splitting or of a threshold kind [Sha79]). At the end of the protocol, the five parties will all know the public output (as prescribed by $\mathcal{F}$), and they can publish it on a public bulletin-board, such as a permissioned blockchain. The actual

process of doing the trade, i.e. which CUSIP numbered shares go to which broker can be done by some clearing agency (it may charge a small fees for its services).

2.1 Improved Liquidity

Since the limit bids and asks are now completely secret (other than what is disclosed from the output of $\mathcal{F}$), the main concern of the traders in submitting large bid and ask goes away. For example, a typical example often cited in the current stock market structure is that not all investors maybe present at the same time, even though they are willing to take opposite sides of the trade. An investor A may want to sell 1000,000 shares of Apple stock today at current prices, whereas another investor B may want to buy 1000,000 shares at current prices, but only tomorrow due to some independent reasons (assuming no new drastic information is generated). In the current system, if a large sell order by A is put into the market, the prices immediately drop (this behavior of the market holds even in the NYSE specialist system [MC00]). In essence, investor A ends up paying the cost of liquidity to other intermediaries who serve as liquidity providers. However, in our system, if investor A is patient for a day or two, or sometimes even a few CPDA sessions, partial information leaked by outputs of $\mathcal{F}$ in different CPDA sessions, will still hide the remaining open orders of A. We model and analyze this behavior of our mechanism more rigorously in Section 4.1.

3 Economic and Game Theory Preliminaries

Giving a detailed presentation of economic theory models and primitives is beyond the scope of this paper, but we point to the excellent teaching notes of Jonathan Levin [Lev06]. Very briefly, consider N resources and M agents. Each agent i has an endowment of resource $j : e_j^i$. Each agent has a utility function (of their endowments) : u_j^i . An economy is $\mathcal{E} = (u^i, \bar{e}^i)_{i \in [M]}$. Now consider a vector of market prices (of resources): $\bar{p}$. Each agent tries to maximize their utility under their budget constraint: $\bar{p} \cdot \bar{x}^i \leq \bar{p} \cdot \bar{e}^i$, where $\bar{x}_j^i$ is the new allocation of resource j to agent i . A *Walrasian equilibrium* of an economy $\mathcal{E}$ consists of (a) a price vector $\bar{p}$, and (b) a redistribution of resources $\{\bar{x}^i\}_{i \in [M]}$, such that each agent maximizes his/her utility and markets clear, i.e. for all resources j we have $\sum_i \bar{x}_j^i = \sum_i e_j^i$.

A feasible reallocation $\{\bar{x}^i\}$ of economy $\mathcal{E}$ is called *Pareto Optimal*, if there is no other feasible allocation $\{\bar{y}^i\}$ such that for all $i \in [M]$: $u^i(\bar{y}^i) \geq u^i(\bar{x}^i)$, with strict inequality for some $i \in [M]$. The first Welfare Theorem [AD54] states that a redistribution of resources that is in Walrasian equilibrium is Pareto optimal.

A normal form Game G consists of a set of players $i \in [M]$, and a set of strategies for each player denoted S_i . Let $S = S_1 \times S_2 \times \dots \times S_M$. Each user has a payoff function $u^i : S \rightarrow \mathbb{R}$. A (mixed) strategy for player i , σ_i is a probability distribution on S_i . A strategy profile $(\sigma_1, \dots, \sigma_M)$ is a *Nash equilibrium* of G if for every i , and every $s_i \in S_i$, $u^i(\sigma_i, \vec{\sigma}_{-i}) \geq u^i(s_i, \vec{\sigma}_{-i})$.

4 Modeling Research Advantage in Rational Expectations Equilibrium

Since double auctions are notoriously difficult to analyze, we will make some simplifying assumptions. We will assume, as in [EO04], that the seller(s) have a supply of stocks to sell, which follows a normal distribution.

Each buyer has a choice of investments which can either be cash, or one of the n risky stocks. Let's say at the beginning of a day when the buyer needs to make an investment choice, it is public knowledge that the value of the k -th stock in the future (or next day), denoted by random variable v_k , will follow a normal distribution with mean $\bar{v}_k$ and precision ρ_k . This, of course, is a base case distribution. Some traders may have proprietary research that could change their future valuation of each stock. Using the terminology of [GS80], we will call this an **observation**, which could have a cost. This observation leads to the trader getting a (noisy) signal about the actual future price v_k . For simplicity, we will assume that such *informed traders* obtain the same signal s_k , and the probability distribution of s_k is itself a normal distribution with mean v_k and precision γ_k conditioned on the future valuation of the stock being v_k . Thus, s_k more or less predicts v_k with precision γ_k (in other words, γ_k can be seen as measuring the historical accuracy of their research about the k -th stock). In fact as observed in [EO04], conditioned on the signal s_k , the distribution of v_k can be calculated by Bayes rule to be a normal distribution with mean $(\bar{v}_k\rho_k + s_k\gamma_k)/(\rho_k + \gamma_k)$ and precision $\rho_k + \gamma_k$.

We will also assume that each buyer (investor) T_j has an exponential utility function with risk-aversion coefficient δ_j . In other words, the buyer's utility function u_j is the following function of wealth w_j : $u_j(w_j) = (1 - e^{-\delta_j w_j})/\delta_j$. It is well known that such utility functions are CARA (constant absolute risk aversion), and we will use this fact later. It is safe to assume that big institutions are more risk-neutral, meaning their δ is close to zero, whereas individual investors will tend to have a large δ .

For any buyer T_j with risk-aversion coefficient δ_j , and initial wealth w_j , suppose T_j buys $x_{j,k}$ quantity of the k -th stock at price p_k . Then, the cash d_j it has left is $w_j - \sum x_{j,k}p_k$. Thus, their future wealth Ψ_j is $d_j + \sum x_{j,k}v_k$, which is same as

$$w_j + \sum_k (v_k - p_k) \cdot x_{j,k}$$

Now, suppose that the buyers know the price they have to pay for the k -th stock, e.g. price in rational expectations equilibrium. Then, their future utility can be maximized w.r.t. each of the quantities $x_{j,k}$. For the buyers who receive a signal for the k -th stock, we can also condition on the signal being s_k . In either case, i.e. whether they receive a signal or not, the distribution of v_k remains normal with a mean and precision known to the buyer. As mentioned earlier, if they do receive a signal s_k , then the conditional distribution of v_k is still normal with mean $\bar{v}_k\rho_k + s_k\gamma_k/(\rho_k + \gamma_k)$ and precision $\rho_k + \gamma_k$.

It can be shown (see e.g. [EO04]) that if the utility function of T_j is of the above exponential form with risk-aversion coefficient δ_j , then it suffices to

maximize $E[\Psi_j] - (\delta_j/2) \cdot \text{Var}[\Psi_j]$ (called *mean-variance*). It is not difficult to see that maximum is achieved at

$$x_{j,k} = \frac{\bar{v}_{j,k} - p_k}{\delta_j(\rho_{j,k})^{-1}} \quad (1)$$

where $\bar{v}_{j,k}$ and $\rho_{j,k}$ are the (conditional) expectation and variance resp. of v_k (conditioned on the signal s_k for the buyers who receive signal s_k).

With this analysis, [EO04] conclude that if *all* traders receive the signal, and assuming all of them have the same δ , and further if there is a per capita supply y_k of the k -th stock, then since in rational expectations equilibrium supply equals demand, we get that

$$p_k = \frac{\bar{v}_k \cdot \rho_k + s_k \cdot \gamma_k - \delta \cdot y_k}{\rho_k + \gamma_k}. \quad (2)$$

Thus, the expected return is $E[v_k - p_k] = \delta \cdot \bar{y}_k / (\rho_k + \gamma_k)$. However, the situation gets more interesting if only a fraction μ_k of traders get the signal for the k -th stock. In the rational expectations equilibrium model, the uninformed buyers (i.e. those who do not observe, and hence do not receive the signal) can conjecture the signal from the price p_k as long as the distribution of the signal is public knowledge. In particular, [EO04] show that in case $\mu_k > 0$ then conditioned on the price p_k , the uninformed buyers can compute a random variable θ which is distributed normally with mean v_k (not to be confused with $\bar{v}_k$) and precision $\rho_{\theta,k}$ given by

$$\left[\gamma_k^{-1} + \left(\frac{\delta}{\mu_k \gamma_k} \right)^2 \right]^{-1} \quad (3)$$

Note that if δ is zero, as is the case for risk-neutral buyers, then this is the same distribution as that of the signal, and hence the price completely reveals whatever advantage the informed buyers get from the signal. On the other hand, if δ is non-zero, then the price is only partially revealing to the uninformed in equilibrium, and there remains an advantage to the informed. Interestingly, the expected return *per share* for each investor (informed or uninformed), now is $\delta \cdot \bar{y}_k / (\rho_k + \mu_k \gamma_k + (1 - \mu_k) \rho_{\theta,k})$ (see [EO04, Prop. 2]). The total return of informed is higher, as they get more or less shares based on the signal. One interesting observation is that with $\mu_k < 1$, i.e. less traders are informed, the return (per share) increases, as traders demand more return as they are less informed, which is a good thing for the fraction who are informed. If $\mu_k = 0$, then we have the most expectation of return, as everyone demands a premium now to hold the stock about which no one has any observations. This would indeed be the case if the incentive of the informed was removed, and their signal was revealed to all⁴. To elaborate, consider the situation where some fraction of traders did research and got a signal s_k , but then signal was leaked to all, so

⁴ In an open bid setting, let's say the signal was not directly leaked to informed traders, but the bids at each price were leaked. Then, by (1), from the demand of informed, an uninformed trader can easily calculate the signal s_k .

essentially μ_k becomes one. As noted above, this is the situation where there is least return for all, or alternatively seen as traders demanding least return (in competition with holding cash). However, the return is actually worse for the traders who did research as they had a fixed cost of research as well. Thus, this would force no one to do research (i.e. if their signal was given to all). We can also check from above that when μ_k is thus forced to become zero, the return required off the stock goes up considerably (precisely, $\delta \cdot \bar{y}_k / \rho_k$ for $\mu_k = 0$ compared to $\delta \cdot \bar{y}_k / (\rho_k + \gamma_k)$ for $\mu_k = 1$).

4.1 Effects of Micro-Structure of Price Discovery on Asset Returns

While the analysis above was done assuming rational expectations equilibrium without regard to the mechanism which may be required to achieve such an equilibrium, we now focus on practical mechanisms and analyze the effects on returns for both informed and uninformed traders.

The main goal is to show that sealed-bid and sealed-ask clearing price double auctions of the particular form as described above have the property that (a) the average return of an informed investor is higher than in the basic rational expectations equilibrium analysis, and (b) if all investors are equally informed then the clearing price rapidly converges to the rational expectations equilibrium price.

Actually, the property (b) has been well-studied in the individual private belief model, and it is shown in [SW89] that if the number of traders is large then they converge to bidding truthfully, i.e. in accordance with their private valuation. Also see [KN04], where it is shown that discrete bidding strategies (as opposed to continuous bidding functions) make the clearing price auction efficient.

So focusing on (a), we first remark that rewarding an investor who has put costly effort into researching a stock is not just important from an equilibrium perspective, but one can reasonably argue that such an equilibrium tends to incentivize research and hence more efficient utilization of capital. Further, the uninformed also continue to have positive returns, since the signal that the informed receive is not absolutely precise. Thus, the uninformed remain in the game, and hence provide liquidity. Finally, when the information becomes public, e.g. after an earnings report, the playing field is leveled and the uninformed get even higher returns, i.e. are more competitive with informed players.

Price Discovery under Common Public Information We first focus on the case where no buyer (or seller) receives a signal (and it is common knowledge), and the distribution of the future price v is public knowledge to be a normal distribution with mean $\bar{v}$ and precision ρ . All buyers and sellers are assumed to have the same risk-aversion coefficient δ . For simplicity of exposition, we will assume that there are exactly n buyers and exactly n sellers. Moreover, each investor has only risk-free cash as an alternative. The buyers will be assumed to own no stock, and the sellers will be assumed to own on average per-capita stock

of z (random variable) shares (say, z_j for the j -th seller). Also, for simplicity, we will assume that the CPDA has infinite precision, i.e. bids, asks and the clearing price can be arbitrary real numbers.

We claim that the following strategies for the buyers and sellers converges to a (bayesian) Nash equilibrium, as n tends to infinity.

- For each price $p < \bar{v}$, each buyer puts in an order to buy ρ/δ shares of stock at limit price p .
- At each price $p > \bar{v} - z_j \cdot \delta/\rho$, j -th seller puts in an order to sell ρ/δ shares of stock at limit price p . Also, for prices $p < \bar{v} - z_j \cdot \delta/\rho$, the j -th seller becomes a buyer and puts in an order to buy ρ/δ shares at limit price p .

Theorem 1. *Let there be n buyers and n sellers. Then, as n tends to infinity, the above strategies constitute a bayesian Nash equilibrium for the CPDA mechanism described in Section 2. The strategies are also truthful⁵.*

Proof. If there are n buyers and n sellers with these strategies⁶, the net demand for the stock at price p is

$$B(p) - S(p) = n \cdot (\bar{v} - p) \cdot \rho/\delta - (n \cdot z - n \cdot (\bar{v} - p) \cdot \rho/\delta).$$

Thus, the clearing price p^* is $p^* = \bar{v} - \rho^{-1}\delta \cdot z/2$. At this price, the j -th buyer gets exactly $z/2$ shares. Note that since the buyer's utility function is CARA (constant absolute risk aversion), and all random variables follow normal distribution, the buyer's utility is an increasing monotonic function of the standard mean-variance expression, i.e. $E[\Psi_j] - (\delta/2) \cdot \text{Var}[\Psi_j]$, where, as in Section 4, Ψ_j is j -th buyer's future wealth. Then, ignoring the cash position of the j -th buyer, the above mean-variance expression is easily calculated to be $z^2\rho^{-1}\delta/8$.

We now show that this is “almost” a bayesian Nash equilibrium, in the sense that if n tends to infinity, then it converges to a Nash equilibrium. For finite n , an investor can improve upon his “equilibrium” strategy by only a factor of $1/(1 + 1/2n)$. This is commensurate with [SW89] where they only considered CPDA with each agent bidding or offering exactly one item.

Suppose, all agents but one fix their strategy to be as described above. We will only consider the case where the exceptional agent is a buyer (the other case is symmetric and analyzed similarly). The exceptional agent choses an alternate strategy of demanding a cumulative $D(p)$ stock at price p . For comparison, the “equilibrium” strategy above sets $D(p) = (\bar{v} - p) \cdot \rho/\delta$, for $p < \bar{v}$, and zero everywhere else. The total demand for the stock at price p is now

$$\begin{aligned} B(p) - S(p) &= n \cdot (\bar{v} - p) \cdot \rho/\delta - (\bar{v} - p) \cdot \rho/\delta + D(p) - (n \cdot z - n \cdot (\bar{v} - p) \cdot \rho/\delta) \\ &= (2n - 1) \cdot (\bar{v} - p) \cdot \rho/\delta + D(p) - n \cdot z \\ &= (2n - 1) \cdot \rho/\delta \cdot (\bar{v} - n/(2n - 1) \cdot z \cdot \rho^{-1}\delta - (p - D(p)\rho^{-1}\delta/(2n - 1))) \end{aligned}$$

⁵ A strategy for a player in the auction is called truthful, if the size of bids and asks (in the strategy) at each price point are exactly the demand/supply function based on the player's valuation in a Walrasian equilibrium (for instance given in equation (1)).

⁶ We will also assume that the number of buyers is large enough that budget constraints of individual buyers does not come into play.

Thus, the clearing price p^{**} satisfies

$$p^{**} - D(p^{**}) \cdot \rho^{-1} \delta / (2n - 1) = \bar{v} - n / (2n - 1) \cdot z \cdot \rho^{-1} \delta \quad (4)$$

Note that as n tends to infinity, this is same as the clearing price p^* computed earlier where all agents used the prescribed strategy. We need to compare if the exceptional buyer's (say, j -th buyer) utility can be improved upon with the alternative strategy $D(p)$. At the above price p^{**} , the j -th buyer gets full allotment $D(p^{**})$, and hence the above mean-variance expression difference from "equilibrium" expression is (after some simple manipulation and using (4))

$$(z \cdot 2n / (2n - 1) - D(p^{**}) \cdot (2n + 1) / (2n - 1)) \cdot \rho^{-1} \delta / 2 \cdot D(p^{**}) - z^2 \cdot \rho^{-1} \delta / 8$$

Taking the derivative of this w.r.t. $D(p^{**})$ shows that the maximum is achieved at $D(p^{**}) = z / 2 \cdot 2n / (2n + 1)$. Thus, with n tending to infinity, this approaches the equilibrium quantity, and hence we obtain a Nash equilibrium in the limit. For finite n , it can be shown that the exceptional buyer can follow the strategy $D(p)$ of bidding $(2n - 1) / 2n \cdot \rho / \delta$ shares at each limit price p with $p < (2n - 1) / 2n \cdot \bar{v}$ to get $D(p^{**}) = z / 2 \cdot 2n / (2n + 1)$. This only differs from the limiting Nash equilibrium strategy by a factor of $1 / (1 + 1/2n)$.

Price Discovery under Mixed Private and Public Information The situation gets interesting when only a fraction of investors receive a pointed signal about the future stock price(s). Recall that in such a scenario, in the rational expectations equilibrium setting above (section 4), the price is only partially revealing to the uninformed buyers. We will now illustrate that in the CPDA considered in this work, that the informed are even more incentivized to gather information than suggested by the rational expectations equilibrium and are incentivized even if they are risk neutral. Moreover, the uninformed also continue to get a positive return on their investment.

Again, for simplicity of exposition we will assume that all the n buyers are uninformed and own no stock, n_1 of the sellers are uninformed, and $n_2 = n - n_1$ sellers are informed and get a signal s from a normal distribution with mean v (the actual future price) and precision γ . The sellers, on average, own per capita stock of z . We will now show that as n tends to infinity, the following strategies form a bayesian Nash equilibrium.

- The j -th uninformed seller, who own z_j stock, behaves as in previous subsection, i.e. at each price $p > \bar{v} - z_j \cdot \delta / \rho$, j -th seller puts in an order to sell ρ / δ shares of stock at limit price p . Also, for prices $p < \bar{v} - z_j \cdot \delta / \rho$, the j -th uninformed seller becomes a buyer and puts in an order to buy ρ / δ shares at limit price p .
- The informed, who receive a signal s , can be shown to believe that the future price follows a conditional distribution that is a normal distribution with mean $\bar{v}_{|s} = (\rho \cdot \bar{v} + \gamma \cdot s) / (\rho + \gamma)$ and precision $\rho + \gamma$ [EO04]. Thus, the informed seller, who owns z_j of stock, does the following: for each price $p > \bar{v}_{|s} - z_j \cdot \delta / (\rho + \gamma)$, the informed seller puts in an order to sell $(\rho + \gamma) / \delta$ shares of stock at limit

- price p . Also, for $p < \bar{v}_s - z_j \cdot \delta / (\rho + \gamma)$, this agent becomes a buyer, and puts in an order to buy $(\rho + \gamma) / \delta$ shares of stock at limit price p .
- The n uninformed buyers, who own no stock, do the following: for each price $p < \bar{v}$, each uninformed buyer puts in an order to buy ρ / δ shares of stock at limit price p .

Theorem 2. *As n tends to infinity, and n_2 remaining finite, the above strategies constitute a bayesian NASH equilibrium for the CPDA mechanism described in Section 2. The strategies are also truthful.*

The proof of this theorem can be found in Appendix A.

Note that when $s < \bar{v} + n / (n - n_1) \cdot z \delta / \gamma$, we have $p^* < \bar{v}$, and the uninformed buyers also get an allocation, and their future utility is positive. Thus, even though the informed get a pointed signal from their research, as long as it is not grossly lopsided, the uninformed get a price which is below the mean future price, and hence their future utility is positive. We also have the following Corollary.

Corollary 1. *When $\delta = 0$, i.e. traders are risk neutral, all traders get zero average return in the rational expectations equilibrium, whereas in the above CPDA, the informed get a positive return that is better than any return of the uninformed.*

The proof of this corollary can be found in Appendix 1.

Does the Volume of the CPDA reveal information? While in the rational expectations equilibrium model, Easley and O’Hara [EO04] show that the equilibrium price reveals only partial information about the signal that the informed receive, their model does not disclose the volume of trade that happens at equilibrium. However, market mechanisms such as the one described in Section 2 do reveal both the clearing price and the volume of trade in each CPDA. The question naturally arises as to whether the additional volume information reveal more information to the uninformed as can be garnered from just the price. First of all, in the periodic CPDA version of the stock market, each individual round reveals additional information to the uninformed, since at the end of each round the uninformed have access to the clearing price, and also the volume of trade in that round. If one runs enough rounds of CPDA, it is expected to reach an equilibrium similar to the one modeled in [EO04] by Walras’ intuitive tatonnement.

In this section, we take a finer look at the model in the previous subsection 4.1, by considering the distribution of the stock held by investors in determining the volume of trade in an individual round of CPDA. Previously, we had assumed that on average the sellers hold per-capita stock of z shares (a random variable). We can model this better by assuming that the informed sellers individually own stock the size of which is sampled from a normal distribution with mean z_2 and variance η_2 , and similarly the uninformed own stock which is sampled from a normal distribution with mean z_1 and variance η_1 .

Theorem 3. *If the CPDA ideal functionality $\mathcal{F}_{\text{CPDA}}$ reveals both clearing price and volume, the signal s observed by informed traders is only partially revealed.*

The proof of this theorem can be found in Appendix A.

5 Implications and Conclusion

In this work we have considered the possibility of replacing full trust in a single auctioneer in a stock market by a collection of auctioneers employing a cryptographic protocol. The cryptographic protocol, known as secure multi-party computation, is well studied in the field of cryptography, and guarantees the privacy of the investors' bids and asks as long as not all of the auctioneers collude. Thus, even if a single auctioneer refuses to join in a collusion, the other colluding auctioneers learn nothing other than the legitimate output of the auction. The output of the auction is the clearing price and the volume. We have argued, using economic and game-theoretic modeling, that not only does this enhanced trust in the privacy of the auction incentivize independent research in the underlying stock, it also improves liquidity in the market as investors can leave large limit orders on the order books without the concern of their private information leaking out. The enhanced trust may also help reverse the trend towards NASDAQ style open order books, as that trend is possibly a result of skepticism in the benevolence of specialist auctioneers. To quote T. Cason and D. Friedman [CF97], "the benevolent auctioneer is at best an endangered species" (for empirical studies on negative specialist behavior, see [MC00,Ben06]). The need for sealed bid-ask mechanisms is further amplified by our result, enhancing an earlier result of [SW89], that shows the sealed bid-ask clearing price double auction to rapidly converge (as the number of investors participating in the CPDA become large) to truthful behavior of investors in the usual CAPM (capital assets pricing) model extended by the Easley-O'Hara model of private information [EO04].

From a real world impact perspective, one can even have an independent regulator as one of the auctioneers, and thus practically removing any chance of full collusion among the auctioneers. The secret randomness used in the protocol can be saved in a block chain to be revealed later for public audit, thus enhancing trust in the system. The proposed mechanism and cryptographic protocol will not only incentivize independent research in the stock, thus enabling a better information structure around the stock, but also circumvent front running and obviate the need for dark pools.

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A Proofs of Nash Equilibrium Theorems

A.1 Proof of Theorem 2

The following strategies form a bayesian Nash equilibrium (as shown in Theorem 2).

- The j -th uninformed seller, who own z_j stock, behaves as in previous subsection, i.e. at each price $p > \bar{v} - z_j \cdot \delta / \rho$, j -th seller puts in an order to sell ρ / δ shares of stock at limit price p . Also, for prices $p < \bar{v} - z_j \cdot \delta / \rho$, the j -th uninformed seller becomes a buyer and puts in an order to buy ρ / δ shares at limit price p .
- The informed, who receive a signal s , can be shown to believe that the future price follows a conditional distribution that is a normal distribution with mean $\bar{v}_{|s} = (\rho \cdot \bar{v} + \gamma \cdot s) / (\rho + \gamma)$ and precision $\rho + \gamma$ [EO04]. Thus, the informed seller, who owns z_j of stock, does the following: for each price $p > \bar{v}_{|s} - z_j \cdot \delta / (\rho + \gamma)$, the informed seller puts in an order to sell $(\rho + \gamma) / \delta$ shares of stock at limit price p . Also, for $p < \bar{v}_{|s} - z_j \cdot \delta / (\rho + \gamma)$, this agent becomes a buyer, and puts in an order to buy $(\rho + \gamma) / \delta$ shares of stock at limit price p .
- The n uninformed buyers, who own no stock, do the following: for each price $p < \bar{v}$, each uninformed buyer puts in an order to buy ρ / δ shares of stock at limit price p .

Theorem 2 (repeated) As n tends to infinity, and n_2 remaining finite, the above strategies constitute a bayesian NASH equilibrium for the CPDA mechanism described in Section 2. The strategies are also truthful.

Proof. Consider the case that $s > \bar{v}$ (the other case is similarly handled). In that case, the net demand for the stock $B(p) - S(p)$ at price p , for $\bar{v} \leq p < \bar{v}_{|s}$, is given by

$$- (n \cdot z - n_1 \cdot (\bar{v} - p) \cdot \rho / \delta - (n - n_1) \cdot ((\rho \cdot \bar{v} + \gamma \cdot s) / (\rho + \gamma) - p) \cdot (\rho + \gamma) / \delta),$$

and is given by

$$n \cdot (\bar{v} - p) \cdot \rho / \delta - (n \cdot z - n_1 \cdot (\bar{v} - p) \cdot \rho / \delta - (n - n_1) \cdot ((\rho \cdot \bar{v} + \gamma \cdot s) / (\rho + \gamma) - p) \cdot (\rho + \gamma) / \delta),$$

for $p < \bar{v}$. Thus, if all agents follow the prescribed strategies, the clearing price is

$$p^* = \frac{(n - n_1) \cdot \gamma \cdot s + n \cdot \rho \cdot \bar{v} - n \cdot \delta \cdot z}{n \cdot \rho + (n - n_1) \cdot \gamma}, \quad (5)$$

if $s > \bar{v} + n/(n - n_1) \cdot z\delta/\gamma$, else

$$p^* = \frac{(n - n_1) \cdot \gamma \cdot s + 2n \cdot \rho \cdot \bar{v} - n \cdot \delta \cdot z}{2n \cdot \rho + (n - n_1) \cdot \gamma}. \quad (6)$$

In the latter case, $p^* < \bar{v}$ (and $s < \bar{v} + n/(n - n_1) \cdot z\delta/\gamma$), and even the uninformed buyers get an allotment. For simplicity, we will now focus on the this second case. Now, given the information, the informed and the uninformed have a different future distribution of the stocks price. While for the uninformed, the future price distribution remains normal with mean $\bar{v}$ and precision ρ , using Bayes principle, it can be seen that having received the signal s , the informed's conditional distribution of the future price of the stock is normal with mean

$$\bar{v}_{|s} = \frac{\rho\bar{v} + \gamma \cdot s}{\rho + \gamma}$$

and precision $\rho + \gamma$. Thus, the mean-variance expression of the uninformed's utility, having obtained their allocation or sale at price p^* above (5), and taking into account their previous ownership of stock, is given by

$$\frac{(\bar{v} - p^*)^2}{2\delta\rho^{-1}}.$$

The informed's ownership w after the CPDA will be

$$(\bar{v}_{|s} - p^*) \cdot (\rho + \gamma)/\delta. \quad (7)$$

So, the informed's mean-variance expression is given by

$$\begin{aligned} & \mathbb{E}[(v - p^*) \cdot w \mid s] - \delta/2 \cdot \text{Var}[(v - p^*) \cdot w \mid s] \\ &= (\bar{v}_{|s} - p^*) \cdot w - \delta/2 \cdot (\rho + \gamma) \cdot w^2 \\ &= (\bar{v}_{|s} - p^*) \cdot (\bar{v}_{|s} - p^*) \cdot (\rho + \gamma)/\delta - \delta/2 \cdot (\rho + \gamma)^{-1} \cdot (\bar{v}_{|s} - p^*)^2 \cdot (\rho + \gamma)^2/\delta^2 \\ &= 1/2 \cdot (\bar{v}_{|s} - p^*)^2 \cdot (\rho + \gamma)/\delta \\ &= 1/2 \cdot w^2 \cdot (\rho + \gamma)^{-1}\delta \end{aligned} \quad (8)$$

As in Section 4.1, now we consider the case where all the strategies are fixed, except for one informed seller/buyer, who instead uses a different ownership (post-CPDA) function $D(p)$. In other words, if he/she already owns z_j of stock, then his/her demand is $D(p) - z_j$. In that case $B(p) - S(p)$, for $\bar{v} \leq p$ is given by

$$\begin{aligned} B(p) - S(p) &= n \cdot (\bar{v} - p) \cdot \rho/\delta + D(p) \\ &\quad - (n \cdot z - n_1 \cdot (\bar{v} - p) \cdot \rho/\delta - (n - n_1 - 1) \cdot ((\rho \cdot \bar{v} + \gamma \cdot s)/(\rho + \gamma) - p) \cdot (\rho + \gamma)/\delta) \\ &= (n - n_1 - 1) \cdot (\rho \cdot \bar{v} + \gamma \cdot s)/\delta - (n \cdot z - (n + n_1) \cdot \bar{v} \cdot \rho/\delta) \\ &\quad - p \cdot \rho/\delta \cdot (2n - 1) - p \cdot \gamma/\delta \cdot (n - n_1 - 1) + D(p), \end{aligned}$$

and hence the clearing price p^{**} satisfies

$$p^{**} \cdot (2n-1) + p^{**} \cdot (n-n_1-1) \cdot \rho^{-1} \gamma - D(p^{**}) \cdot \rho^{-1} \delta = (2n-1) \cdot \bar{v} + (n-n_1-1) \cdot \rho^{-1} \gamma \cdot s - n \cdot z \cdot \rho^{-1} \delta \quad (9)$$

Now, the mean-variance expression of this exceptional informed investor is given by

$$(\bar{v}_{|s} - p^{**}) \cdot D(p^{**}) - \delta/2 \cdot (\rho + \gamma)^{-1} \cdot D(p^{**})^2.$$

Using (9) and taking the derivative of the above w.r.t. $D(p^{**})$, we find that the maximum is obtained at $D(p^{**})$ satisfying

$$\begin{aligned} \bar{v}_{|s} - \frac{(2n-1) \cdot \rho \cdot \bar{v} + (n-n_1-1) \cdot \gamma \cdot s - n \cdot \delta \cdot z}{(2n-1) \cdot \rho + (n-n_1-1) \cdot \gamma} \\ = D(p^{**}) \cdot \left(\frac{2\delta}{(2n-1) \cdot \rho + (n-n_1-1) \cdot \gamma} + \frac{\delta}{\rho + \gamma} \right) \end{aligned}$$

With n tending to infinity, even if $n_2 = (n - n_1)$ remains finite (i.e. the number of informed investors remains finite or tiny), the above gives

$$D(p^{**}) = \left(\bar{v}_{|s} - \frac{2n \cdot \rho \cdot \bar{v} + (n - n_1) \cdot \gamma \cdot s - n \cdot \delta \cdot z}{2n \cdot \rho + (n - n_1) \cdot \gamma} \right) \cdot (\rho + \gamma) / \delta,$$

which is same as informed's stock ownership at equilibrium given by (7) noting that equilibrium clearing price p^* is given by (5). Then, by (8), the informed's mean-variance expression is also the same as in equilibrium. Thus in the limit, when n tends to infinite, and even if the number n_2 of informed investors remains finite or tiny, the above set of strategies constitute a bayesian Nash equilibrium.

Corollary1 (repeated) When $\delta = 0$, informed investors get zero average return in the rational expectations equilibrium, whereas they get a positive return that is better than any positive return of the informed in the above CPDA.

Proof. The mean-variance expression of the future utility of the informed, as calculated above in the proof, is

$$\begin{aligned} & 1/2 \cdot w^2 \cdot (\rho + \gamma)^{-1} \delta \\ &= 1/2 \cdot ((\bar{v}_{|s} - p^*) \cdot (\rho + \gamma) / \delta)^2 \cdot (\rho + \gamma)^{-1} \delta \\ &= 1/2 \cdot \left(\left(\bar{v}_{|s} - \frac{(n-n_1) \cdot \gamma \cdot s + 2n \cdot \rho \cdot \bar{v} - n \cdot z \cdot \delta}{2n \cdot \rho + (n-n_1) \cdot \gamma} \right) \cdot (\rho + \gamma) / \delta \right)^2 \cdot (\rho + \gamma)^{-1} \delta \\ &= 1/2 \cdot \left(\left(\frac{\rho \cdot \bar{v} + \gamma \cdot s}{\rho + \gamma} - \frac{(n-n_1) \cdot \gamma \cdot s + 2n \cdot \rho \cdot \bar{v} - n \cdot z \cdot \delta}{2n \cdot \rho + (n-n_1) \cdot \gamma} \right) \cdot (\rho + \gamma) / \delta \right)^2 \cdot (\rho + \gamma)^{-1} \delta \\ &= 1/2 \cdot \left(\rho \cdot \bar{v} + \gamma \cdot s - \frac{(n-n_1) \cdot \gamma \cdot s + 2n \cdot \rho \cdot \bar{v} - n \cdot z \cdot \delta}{2n \cdot \rho + (n-n_1) \cdot \gamma} \cdot (\rho + \gamma) \right)^2 \cdot (\rho + \gamma)^{-1} \delta^{-1} \end{aligned}$$

The clearing price above is

$$p^* = \frac{(n - n_1) \cdot \gamma \cdot s + 2n \cdot \rho \cdot \bar{v} - n \cdot \delta \cdot z}{2n \cdot \rho + (n - n_1) \cdot \gamma}.$$

In contrast, [EO04] show that the rational expectations equilibrium price q^* is

$$q^* = \frac{((n - n_1) \cdot \gamma + (n + n_1) \cdot \rho_\theta) \cdot s + 2n \cdot \rho \cdot \bar{v} - n \cdot \delta \cdot z}{2n \cdot \rho + (n + n_1) \cdot \rho_\theta + (n - n_1) \cdot \gamma},$$

where ρ_θ is as given in (3), i.e. $\left[\gamma^{-1} + \left(\frac{2n \cdot \delta}{(n - n_1) \cdot \gamma} \right)^2 \right]^{-1}$. Since we are considering the case where $s > \bar{v}$, i.e. when the signal is higher than the future a priori mean value, we now show that p^* is less than q^* , and hence the informed get a better allocation and hence a better mean-variance expression than in the equilibrium setting. This is not surprising, as in equilibrium, the uninformed also get partial information about the signal as demonstrated in [EO04] (see Section 4). We just show that when δ is zero, i.e. the investors are risk neutral, then $p^* < q^*$ (recall, in the rational expectations equilibrium, if δ is zero, the informed get no advantage for their research). We have, with $\delta = 0$ and hence $\rho_\theta = \gamma$,

$$\begin{aligned} p^* - q^* &= \frac{(n - n_1) \cdot \gamma \cdot s + 2n \cdot \rho \cdot \bar{v}}{2n \cdot \rho + (n - n_1) \cdot \gamma} - \frac{((n - n_1) \cdot \gamma + (n + n_1) \cdot \gamma) \cdot s + 2n \cdot \rho \cdot \bar{v}}{2n \cdot \rho + (n + n_1) \cdot \gamma + (n - n_1) \cdot \gamma} \\ &= \frac{(n - n_1) \cdot \gamma \cdot s + 2n \cdot \rho \cdot \bar{v}}{2n \cdot \rho + (n - n_1) \cdot \gamma} - \frac{(2n \cdot \gamma) \cdot s + 2n \cdot \rho \cdot \bar{v}}{2n \cdot \rho + 2n \cdot \gamma} \\ &= \frac{(n - n_1) \cdot \gamma \cdot s + 2n \cdot \rho \cdot \bar{v}}{2n \cdot \rho + (n - n_1) \cdot \gamma} - \frac{\gamma \cdot s + \rho \cdot \bar{v}}{\rho + \gamma} \\ &= \frac{((n - n_1) \cdot \gamma \cdot s + 2n \cdot \rho \cdot \bar{v}) \cdot (\rho + \gamma) - (\gamma \cdot s + \rho \cdot \bar{v}) \cdot (2n \cdot \rho + (n - n_1) \cdot \gamma)}{(2n \cdot \rho + (n - n_1) \cdot \gamma) \cdot (\rho + \gamma)} \\ &= \frac{(n + n_1) \cdot \gamma \rho \cdot (\bar{v} - s)}{(2n \cdot \rho + (n - n_1) \cdot \gamma) \cdot (\rho + \gamma)} \\ &< 0 \end{aligned}$$

A.2 Proof Under Price and Volume Disclosure

Theorem 3 (repeated) If the CPDA ideal functionality $\mathcal{F}_{\text{CPDA}}$ reveals both clearing price and volume, the signal s observed by informed traders is only partially revealed.

Proof. The volume of the CPDA is decided by individual ownership z_j , in the sense that if for the j -th informed seller, $p < \bar{v}_{|s} - z_j \cdot \delta / (\rho + \gamma)$, then the j -th informed seller becomes a buyer. Now, we look at the clearing price, and determine which of the informed sellers effectively became buyers as their ownership z_j was too small, i.e. $p^* < \bar{v}_{|s} - z_j \cdot \delta / (\rho + \gamma)$, or $z_j < (\bar{v}_{|s} - p^*) \cdot (\rho + \gamma) / \delta$. Similarly, the uninformed sellers become buyers if $z_j < (\bar{v} - p^*) \cdot \rho / \delta$. Now, the average ownership $\bar{w}$ conditioned on the ownership being less than $\bar{z} + \alpha$, when sampling from

a normal distribution with mean $\bar{w}$ and precision η is $\bar{z} - \frac{1}{\sqrt{2\pi\eta}} \cdot e^{-\eta^2\alpha^2/2}$. Thus, given that there are n_1 uninformed sellers, and $(n - n_1)$ informed sellers, the average ownership $\bar{w}_b$ of sellers-turned-buyers is (assuming both $(\bar{v}_{|s} - p^*) \cdot (\rho + \gamma) / \delta$ and $(\bar{v} - p^*) \cdot \rho / \delta$ is more than z_1 and z_2 — other cases are handled similarly)

$$\frac{n_1}{n} z_1 + \frac{n - n_1}{n} z_2 - \frac{n_1}{n \cdot \sqrt{2\pi\eta_1}} \cdot e^{-\eta_1^2((\bar{v} - p^*) \cdot \rho / \delta - z_1)^2/2} - \frac{n - n_1}{n \cdot \sqrt{2\pi\eta_2}} \cdot e^{-\eta_2^2((\bar{v}_{|s} - p^*) \cdot (\rho + \gamma) / \delta - z_2)^2/2}$$

Similarly, the average ownership $\bar{w}$ conditioned on the ownership being more than $\bar{z} + \alpha$ is $\bar{z} + \frac{1}{\sqrt{2\pi\eta}} \cdot e^{-\eta^2\alpha^2/2}$, and hence the average ownership $\bar{w}_s$ of sellers-staying-sellers is

$$\frac{n_1}{n} z_1 + \frac{n - n_1}{n} z_2 + \frac{n_1}{n \cdot \sqrt{2\pi\eta_1}} \cdot e^{-\eta_1^2((\bar{v} - p^*) \cdot \rho / \delta - z_1)^2/2} - \frac{n - n_1}{n \cdot \sqrt{2\pi\eta_2}} \cdot e^{-\eta_2^2((\bar{v}_{|s} - p^*) \cdot (\rho + \gamma) / \delta - z_2)^2/2}$$

Thus, the (clearing) volume $B(p^*)$ ($= S(p^*)$) is

$$\begin{aligned} & n \cdot (\bar{v} - p^*) \cdot \rho / \delta + n_1 \cdot z_1 + (n - n_1) \cdot z_2 - \frac{n_1}{\sqrt{2\pi\eta_1}} \cdot e^{-\eta_1^2((\bar{v} - p^*) \cdot \rho / \delta - z_1)^2/2} \\ & - \frac{n - n_1}{\sqrt{2\pi\eta_2}} \cdot e^{-\eta_2^2((\bar{v}_{|s} - p^*) \cdot (\rho + \gamma) / \delta - z_2)^2/2}. \end{aligned}$$

Thus, while p^* is a linear combination of the secret signal s and the random variable $z = \frac{n_1}{n} z_1 + \frac{n - n_1}{n} z_2$, the volume is a more complex function of s, z_1, z_2 . Thus, if the CPDA result reveals s and the volume, the information about s revealed is only partial (as we have three unknowns and two known expressions).